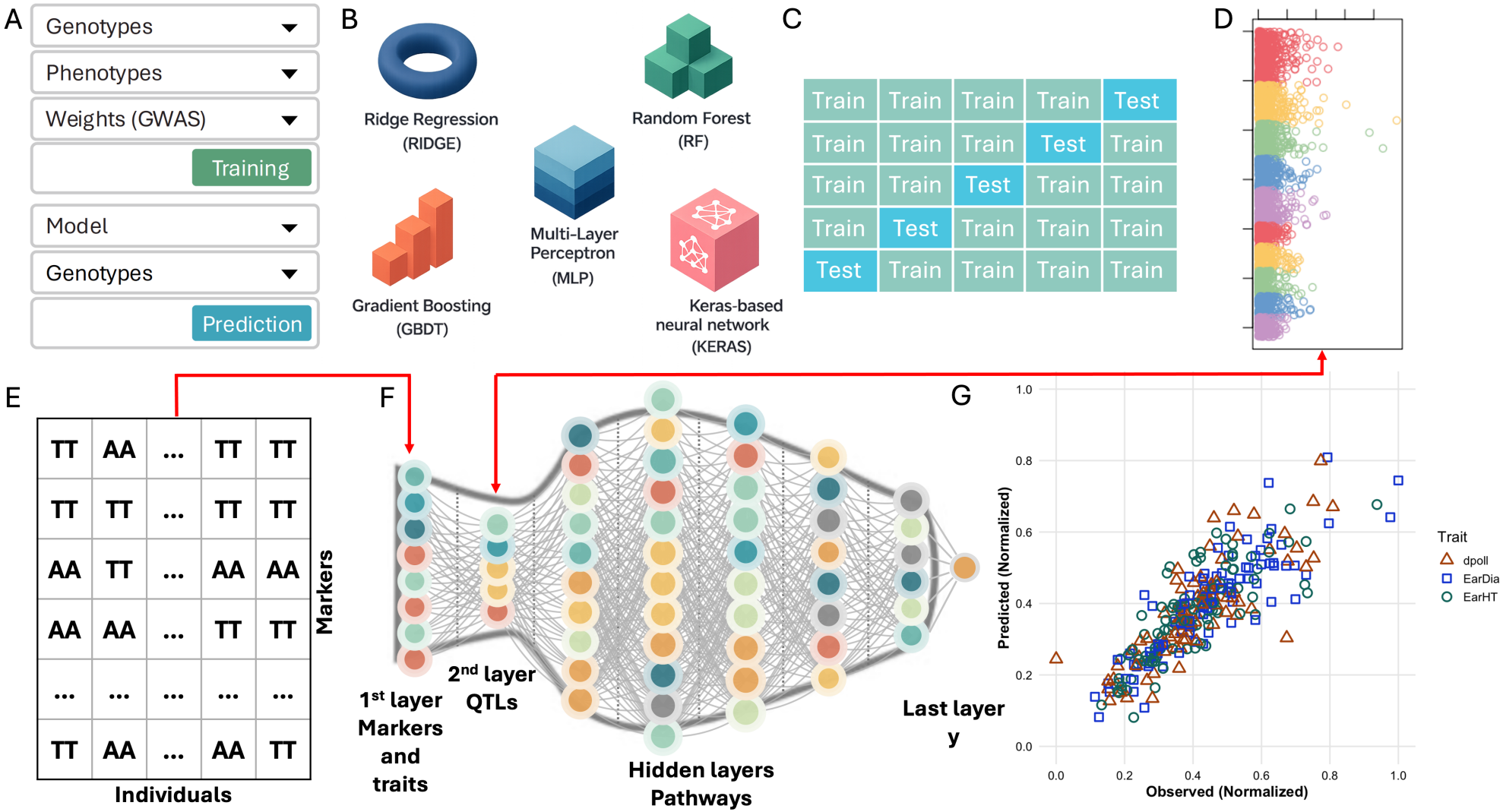
A
Genotypes
Phenotypes
Weights (GWAS)
Training
Model
Genotypes
Prediction
B
Ridge Regression
(RIDGE)
Random Forest
(RF)
Multi-Layer
Perceptron
(MLP)
Gradient Boosting
(GBDT)
Keras-based
neural network
(KERAS)
C
Train Train Train Train Test
Train Train Train Test Train
Train Train Test Train Train
Train Test Train Train Train
Test Train Train Train Train
D
E
TT AA ... TT TT
TT TT ... TT TT
AA TT ... AA AA
AA AA ... TT TT
... ...
TT AA ... AA TT
Markers
Individuals
F
1st layer
Markers
and
traits
2nd layer
QTLs
Hidden layers
Pathways
Last layer
y
G
Predicted (Normalized)
Observed (Normalized)
0.0
0.2
0.4
0.6
0.8
1.0
Trait
dpoll
EarDia
EarHT